

Exercises for Lecture "Theoretical Spectroscopy for systems implying periodic boundary conditions"

Exercise 2: Static IR spectroscopy

based on exercises taken from the [CP2K Homepage](#).

2.1: Geometry optimization, vibrational frequencies and IR spectra

- Optimize the geometry of the given metal-organic framework using the GGA functional PBE as well as using the semi-empirical tight-binding method GFN1-xTB. Compare the inputs named `mof_pbe.inp` and `mof_xtb.inp` as well as the corresponding total electrostatic energies obtained from the two computations.
- Compute corresponding vibrational frequencies by changing the run type from geometry optimization (`RUN_TYPE GEO_OPT`) to vibrational analysis (`RUN_TYPE VIBRATIONAL_ANALYSIS`). Compare and visualize the obtained vibrational frequencies and corresponding intensities using the Fortran code `newspecmatch`. To use `newspecmatch`, insert the following two lines in your `.bashrc` file and source the file:

```
export newspecmatchhome=/zfshome/suphc356/newspecmatch/src
export PATH=$PATH:$newspecmatchhome
```

The option `-plot` enables to plot the spectral function:

```
newspecmatch spectra_1.dat -plot -fscal 0.982 -range 0 3500.
```

With `newspecmatch`, vibrational spectra can also be compared with the norms closer to 1 indicating a good match:

```
newspecmatch spectra_1.dat spectra_2.dat
```
- Scale the obtained xTB frequencies by adjusting the global scaling factor `fscal` and plot the spectra again using the `plot` keyword and `gnuplot spectra.gnup`.